MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE DOD- 011 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, coaxial, tunable frequency range 15,400 to 15,700 MHz, rated peak power output 2.5 kw, with integral magnet

ABSOLUTE RATINGS:

Parameter:	if(surge)	tk	VSWR	Tuner stop torque	T(body)	T(anode)	(Input T bushing)	Cak	tsrv	Alt
Unit:	a	sec	---	in. oz	°C	°C	°C	pF	μs	ft
Maximum:	6	---	1.3:1	20	+71	+150	+175	35	0.150	10,000
Minimum:	---	60	---	---	-54	-54	-54	20	0.06	---

ABSOLUTE RATINGS:

Parameter:	Ef	If	eb	ib	Pi	pi	Du	tpc
Unit:	V	A	v	a	W	kw	---	μs
Maximum:	7	2.5	3,800	3.1	60	12	0.005	0.5
Minimum:	6	1.5	---	---	---	---	0.001	0.2

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Marking:	See note 2
Mounting position:	Any	Magnet isolation:	See note 1
Mounting support:	See figure 1	Weight:	4 pounds (maximum)
Input coupling:	See figure 1	Handling:	See note 1
Output coupling:	See figure 1	Cooling:	Conduction and convection

TEST CONDITIONS:

Parameter:	Ef	ib	Du	tpc	trc
Unit:	V	a	---	μs	μs
Tolerance:	±0.1	---	---	±0.05	±0.01
Test condition 1 (Note 3):	6.3	3	0.005	0.3	0.07
Test condition 2 (Note 4):	6.3	3	.001	.3	.07
Test condition 3 (Note 5):	6.3	3	.005	.3	.07
Test condition 4 (Note 6):	6.3	3	.001	.3	.07

FREQUENCY	
F	MHz
F1	15,412
F2	15,436
F3	15,484
F4	15,508
F5	15,532
F6	15,568
F7	15,592
F8	15,616
F9	15,664
F10	15,688

GENERAL:

Qualification - Required

1/ Replaces AIL P/N 395964-1, Litton type L-5328, and Varian type VMU-1018

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection	-	-	F = F1 thru F10				
1026	Temperature range operating (high)	18	1, 2	T_A = +71°C	MP	---	0.1	%
1026	Temperature range operating (low)	18	1, 2	T_A = -54°C	MP	---	0.1	%
1002	Barometric pressure, reduced	18	1, 2		---	---	---	---
4243	Spurious output ratio	9	-		po	---	-30	dBm
1042	Shock, specified pulse (nonoperating)	7	-	200 G; t = 3 ms	F	---	1.5	MHz
1331	Direct-interelectrode capacitance	8	-		C	20	35	pF
4223	Tuner stop endurance	17	-		---	50	---	in. oz
4027	Temperature coefficient	18	1, 2	T_A = 25° to -54°C to +71° to 25°C	ΔF	---	±3	MHz
4223	Mechanical tuning fatigue	10	-		---	1,500	---	Cycles
4302	Frequency change (warmup)	18	3, 4		ΔF	---	14	MHz
---	Power output (pulse train)	11, 18	1, 2		Δpo	---	0.5	dB
---	Frequency stability (pulse train)	12, 18	1, 2		ΔF	---	1	MHz
---	Frequency stability (filament)	13, 18	3, 4	Ef = 6.3 V ±5%	ΔF	---	1	MHz
	Quality conformance inspection, part I	19	-	F = F1, F4, F7, F10				
1301	Heater current	-	-	Ef = 6.3 V; tk = 60 sec	If	1.5	2.5	A
4303	Warmup time (cathode)	-	-	Ef = 6.3 V; tk = 60 sec (max)	---	---	---	---
4306	Pulse voltage (anode)	-	-		eb	3,400	3,700	v
4250	Power output	18	3, 4		po	2,500	3,500	w
4310	Frequency pulling figure	18	3, 4	VSWR = 1.3:1 (min)	ΔF	---	5	MHz
4311	Frequency pushing figure	18	3, 4	Δfb = ±5%	ΔF	---	±0.5	MHz
4223	Mechanical tuning range	18	3, 4		F	15,400	15,700	MHz

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1 -Continued	19	-	F = F1, F4, F7, F10				
4223	Resettability	18	3, 4		ΔF	---	±5	MHz
4308	RF bandwidth	15, 18	3, 4		F	---	2.5/tpc	MHz
4308	Minor lobe ratio	15, 18	3, 4		Ratio	8	---	dB
4315	Stability	16, 18	3, 4		MP	---	0.1	%
	Quality conformance inspection, part 2	-	-	F = F1, F4, F7, F10				
1031	Operation vibration	7, 18	1, 2		ΔF	---	1	MHz
4223	Tuner torque	-	-	T_A = -54° to +71°C	---	---	20	in. oz
	Quality conformance inspection, part 3	-	-	F = F1, F4, F7, F10				
---	Life test	14	-		t	2,000	---	hrs
---	Life-test end points:							
4250	Power output	18	1, 2		po	2,000	---	w
4306	Pulse voltage (anode)	-	-		eb	3,400	3,800	v
4223	Mechanical tuning range	-	-		F	15,400	15,700	MHz
4223	Resettability	-	-		ΔF	---	±5	MHz
4308	RF bandwidth	15, 18	3, 4		F	---	3.0/tpc	MHz
4308	Minor lobe ratio	15, 18	3, 4		Ratio	6	---	dB
4315	Stability	16, 18	1, 2		MP	---	0.2	%

NOTES:

1. In handling and mounting the tube, care shall be exercised to prevent demagnetization. Ferromagnetic materials or energized magnets shall not be brought within 2 inches of the tube.

2. A calibrated tuner with appropriate dial markings shall be incorporated on the tube. The tuner mechanism shall be capable of setting the tubes frequency within ±5 MHz when the tube is operating at test condition 1 conditions, T_A = 25°C, and when the effects of frequency pushing and pulling are neglected. The tuner is to be rotated in a clockwise direction for each new setting. A tuner calibration chart shall be provided for each of the 10 frequencies specified below:

Channel No.	Frequency
1	F1
2	F2
3	F3
4	F4
5	F5
6	F6
7	F7
8	F8
9	F9
10	F10

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NOTES: - Continued

3. High burst duty. The test shall be performed with a pulse train of 1,667 pulse pairs with a pulse pair spacing of 8 μs, and an interpair interval of 60 μs. This provides a pulse train of 99.96 ms duration. Pulse trains shall be generated every 200 ms.

4. Low burst duty. The test shall be performed with a pulse train of 333 pulse pairs with a pulse pair spacing of 8 μs, and an interpair interval of 60 μs. This provides a pulse train of 19.92 ms duration. Pulse trains shall be generated every 200 ms.

5. High average duty. The test shall be performed with constant repetition rate single pulses spaced 60 μs.

6. Low average duty. The test shall be performed with constant repetition rate single pulses spaced 300 μs.

7. After the performance of the specified test, the tube shall meet the requirements of quality conformance inspection, part 3, life-test end points.

8. The input capacitance shall be measured between the cathode terminal and a mounting plate 1/4 inch (6.35 mm) thick with the minimum cut-out required to flush mount the tube.

9. Spurious emissions shall not have peak power exceeding the specified amount on the tube specification sheet (TSS) within the frequency range of 7.5 GHz and 20 GHz. This limit not applicable within either the designed emission bandwidth or ±5 percent of the center frequency in this TSS.

10. Tubes shall meet the tuner torque requirements after the specified number of cycles.

11. The difference in amplitude between the first and last rf pulse in the pulse train shall not exceed 0.5 dB under the condition that peak current variation of these two pulses be less than 5 percent total.

12. The difference in frequency between the third and last pulse in the pulse train shall not exceed the specified value under the condition that peak current variation of these two pulses be less than 5.0 percent total.

13. When the filament voltage is varied as specified, the measured frequency shall not change more than the specified amount.

14. A life-test cycle shall consist of the following:

Duration	Condition
5 minutes	Off
1 minute	Filament voltage applied
60 minutes	Tube operating

The frequency shall be shifted starting at F1 to the next higher channel each eight operating hours.

15. The tube shall be operated into a transmission line with a VSWR of 1.3:1 minimum, adjusted in phase to produce the maximum spectrum degradation. A suitable spectrum is considered one in which the major lobe has a shape such that its slope does not change sign more than once for power levels greater than the specified dB below its peak.

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NOTES: - Continued

16. Stability shall be measured in terms of the average number of output pulses missing, expressed as a percent of the number of input pulses applied during the period of observation. The missing pulses (MP), due to any causes, are considered to be "missing" if the rf energy is less than 70 percent of the normal energy level. A VSWR = 1.3:1 (minimum) shall be adjusted to that phase producing maximum instability and the missing pulses counted during any consecutive 5-minute interval of a 10-minute test interval.

17. Tuner stops shall be provided to limit rotation of the tuner mechanism. The lower frequency limit shall not be less than 15.375 GHz and the upper frequency limit shall not exceed 15.725 GHz, with the tuner mechanism "set" against the upper or lower stop.

18. This test shall be performed using the requirements of either of the specified test conditions.

19. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

Custodians:
Army - EL
Navy - EC

Review activities:

DSA - ES

User activities:
Army - WC, MU
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

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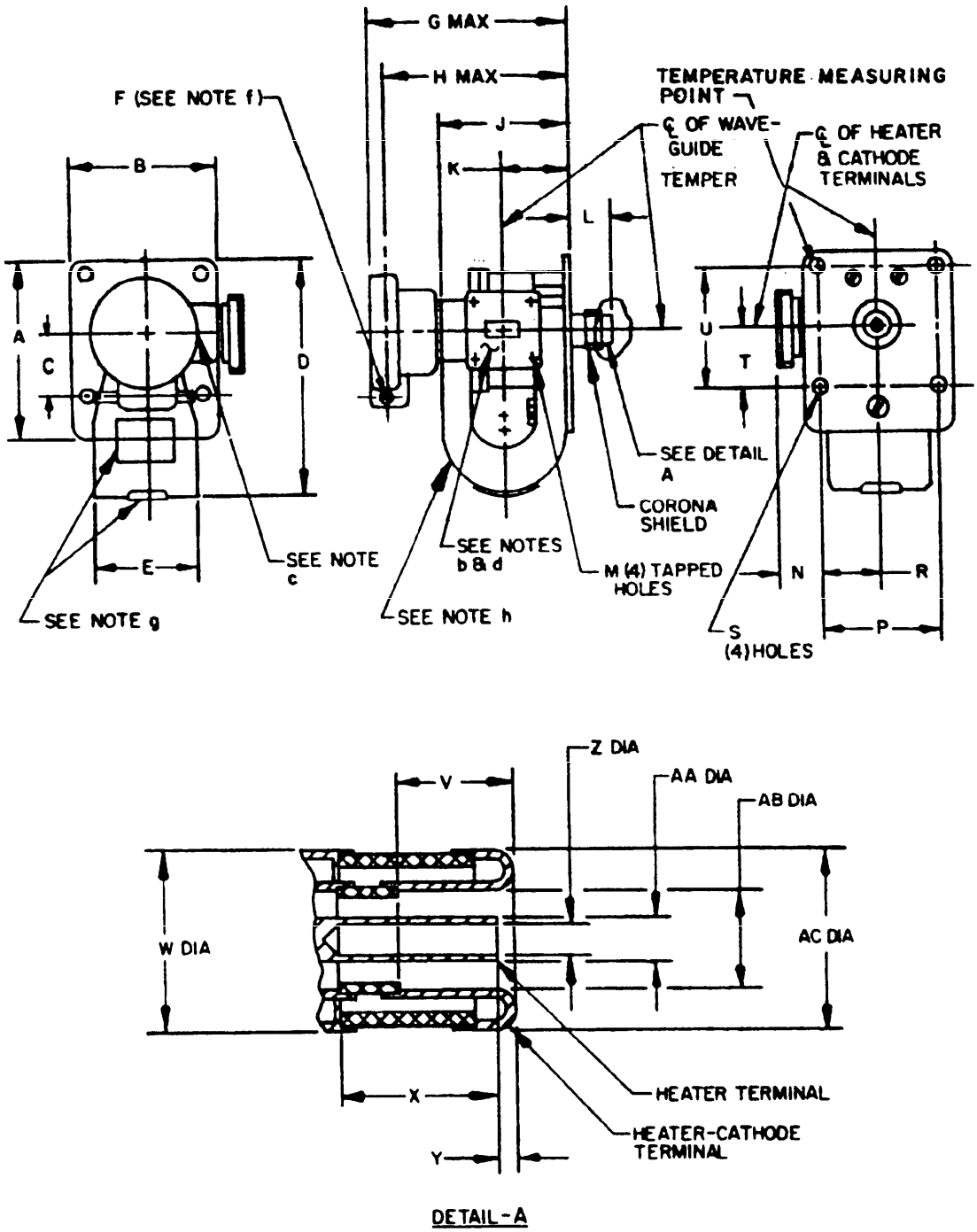


FIGURE 1. Outline drawing of electron tube type DOD-011

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
B	2.490(63.25)	2.510 (63.75)
E		2.000 (50.80)
F	.185 (4.70)	.189 (4.80)
U	1.990(50.55)	2.010 (51.05)
Quality conformance inspection, part 1 (see note 19)		
D		5.000(127.00)
G		3.550 (90.17)
H	3.170(80.52)	3.270 (83.06)
L	.620(15.75)	.740 (18.80)
N	.690(17.53)	.750 (19.05)
P	1.990(50.55)	2.010 (51.05)

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.990(75.95)	3.010 (76.45)
J	2.262(57.45)	2.362 (59.99)
K	1.157(29.39)	1.217 (30.91)
M	6-32 UNC-2B(4 tapped holes)	
R	.995(25.27)	1.005 (25.53)
S	.276 (7.01)	.286 (7.26)
T	.995(25.27)	1.005 (25.53)
V	.312 (7.92)	
X	.312 (7.92)	
Y	.045 (1.14)	.075 (1.91)
AB	.283 (7.19)	.293 (7.44)
Quality conformance inspection, part 3		
Z	.095 (2.41)	.101 (2.57)
AA	.120 (3.05)	.130 (3.30)
AC	.500(12.70)	
Nominal dimensions		
C	1.000(25.40)	
W	.500(12.70)	

NOTES:
a. The mounting flange shall be brass per QQ-B-613, bright alloy finish or aluminum alloy, chemically treated in accordance with MIL-C-5541, class 3.
b. The waveguide flange shall have an electrically conductive finish.
c. The location of tuner calibration indicator specified in note 2 shall be viewed through a sealed window.
d. Waveguide flange mates with UG-541/U or equal modified with clearance mounting holes.
e. Cathode-heater terminals mate with Jettron connector No. 90-050 or equal.
f. Slot is .0425 ± .0020(1.08 ± .05 mm) wide by .125 ± .010(3.18 ± .25 mm) deep.
g. The tube labels shall include tube type, serial number, magnet warning data, and calibration markings.
h. Magnet coated with Silastic or equal.

FIGURE 1. Outline drawing of electron tube type DOD - 011 -Continued.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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